

Saket Gollapudi

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Summary of Qualifications:

- **Technical Skills:** Java, C, C++, Python, PyTorch, System Verilog, SQL, HTML, JavaScript, MATLAB, GIT, PyTorch, ROS, AutoCAD, SolidWorks
- **Soft Skills:** Teaching, Leadership, Critical Thinking, Problem Solving, Public Speaking, Time Management.
- 6+ years of experience in research projects and engineering labs with strong leadership, critical thinking, and teamwork skills
- Passionate about exploring the multidisciplinary applications of computing, focusing on AI, Machine Learning, Robotics.

Education:

UNIVERSITY OF WASHINGTON | SEATTLE, WA

- **Master of Science** ~ Computer Science & Engineering; **Current GPA: 3.88** **EXPECTED GRADUATION: DEC 2026**
 - Courses: Analysis Of Algorithms, Artificial Intelligence, Machine Learning, Robotics, Natural Language Processing, Computer Vision, Deep Learning
- **Bachelor of Science** ~ Computer Engineering; Minor in Aeronautical and Astronautical Engineering; **GPA: 3.94 ~ Magna Cum Laude**
 - **Graduated June 2024; Honors/Awards:** Annual Dean's list 2021, 2023, 2024.
 - Related Coursework: Data Structures/Parallelism, Machine Learning, A.I., Autonomous Robotics, Systems Programming, Security, Distributed Systems, Algorithm Design, Digital Circuits, Orbital/Atmospheric Mechanics, Aerospace Control, Kinematics/Dynamics.

Relevant Experience:

• GRADUATE RESEARCHER | UNIVERSITY OF WASHINGTON | SEATTLE, WA

OCT 2024 - PRESENT

- Researched algorithms for LLM fine-tuning that improve privacy and memory efficiency, with an emphasis on differential privacy and zeroth-order optimization.
- Conducted research under Professor Sewoong Oh, designing experiments to evaluate the performance and scalability of LLM fine-tuning methods.
- First author on paper accepted to the ICML 2026 HILD Workshop: **Understanding Clipping in Zeroth-order Optimization**, investigating the role of per-sample gradient clipping in improving zeroth-order optimization.

• ML/AI ENGINEERING INTERN | T-MOBILE | BELLEVUE, WA

JUN 2025 - SEPT 2025

- Automated critical network anomaly detection, deploying models that directly streamlined operations for customer care teams and reduced manual diagnostic time.
- Engineered ML models to automate network metric analysis, directly influencing executive strategy by presenting data-driven insights to Directors and CNO.
- Built a distributed multihost model, reducing current model training time by 25% while handling 55% more data. Used Tensorflow, Ray, Petastorm on Databricks.

• TEACHING ASSISTANT | UNIVERSITY OF WASHINGTON | SEATTLE, WA

DEC 2022 - DEC 2025

- **CSE 333 - Systems Programming.** Taught C, C++ and basic server behavior to help students build a mini Google search system.
- **CSE 403 - Software Engineering.** Mentored multiple groups to debug/develop various projects from scratch. Help build experience with Git, CI, VC, etc.
- **CSE 311 - Foundations of Computing I.** Taught fundamentals of logic, set theory, and algebraic structures with applications to computing.
- **CSE 312 - Foundations of Computing II.** Taught fundamentals of probability and applications of randomness to computing
- **CSE 446 - Machine Learning.** Taught the fundamentals of ML focusing on design of efficient algorithms that learn from data. Emphasis on algorithmic principles and how to use these tools in practice. Topics include supervised/unsupervised learning, regression and classification, deep learning, and optimization.
- Redesigned curriculum materials and exam content to better align with industry standards for Machine Learning, Systems, and Fundamental CS courses.
- Mentored 200+ students over 3 years, consistently receiving positive feedback for simplifying complex topics and aiding student development/learning.

• AVIONICS SOFTWARE ENGINEERING INTERN | BLUE ORIGIN | KENT, WA

JUN 2023 - SEPT 2023

- Worked under the Director of New Glenn AERS Engineering & Systems to accelerate the verification process for G&C systems aboard New Glenn.
- Enhanced vehicle safety by developing an automated executable testing framework that validated flight software capabilities for the New Glenn launch vehicle.
- Collaborated directly with Chief Engineers to identify critical software defects, presenting root-cause analysis that drove key architectural improvements.

• UNDERGRADUATE RESEARCHER | ROBOT LEARNING LAB – UW | SEATTLE, WA

SEPT 2023 - JUN 2024

- Worked on RACER project to develop a system that provides a geometric and semantic understanding of a vehicle's local environment, autonomously evaluating routes and contingencies, and adapting to behavior of evolving terrain with python and C++, as well as using ROS packages.
- Developed offroad planning module using D* planning algorithm for robot to autonomously find best route if it encounters an obstacle.

• RESEARCH INTERN | AUTONOMOUS FLIGHT SYSTEMS LAB – UW | SEATTLE, WA

SEPT 2019 - JUN 2021

- Assisted in implementing/training a machine learning algorithm using LiDAR to identify objects in dense environments, focusing on human detection via drones.
- 3D-modeled parts for equipment, improved drone structural integrity, and rewired/repared sensors, planned/revised flight paths for efficient/safe testing.
- Collaborated with a team of 6 for 10 hours weekly to train flight software, repair systems, and test autonomous drone operations.

Projects:

- **Robo-Assist (Mar 2024):** Developed an autonomous assistive robot program to manage tasks for high risk pregnancy individuals using Hello Robot's Stretch RE1 robot. Worked in a group setting. Software developed using Ros2, React, Python. Implemented multiple planning, computer vision, and search based algorithms/techniques to read sensor input like camera and LiDAR for main autonomous functionality.
- **Learning Dynamics and Distributional Effects of Pre-training and Reinforcement Learning (Dec 2025):** Investigated dynamics of RL pre-training techniques with Verified Rewards (RLVR) by designing SFT and RL training pipelines for various GPT, Qwen, and Gemma models, utilizing token-level entropy analysis to demonstrate that RL optimization amplifies pre-existing model priors rather than acquiring new domain knowledge.